

Skills Summary

Which Number Is the Slope?

Use this reference while completing your worksheet or when studying.

In $y = mx + b$, position decides the role:

- m multiplies the variable. It is the slope: the change in y for each 1 unit of x . On a graph it is the steepness.
- b is added on. It is the y -intercept: the value of y when x is 0. On a graph it is where the line crosses the vertical axis.
- Test for b : substitute $x = 0$ and see what survives.

Skill 1 — Reading both numbers out of an equation

Rule: the coefficient of the variable is the slope; the lone number is the y -intercept. Swapping them changes the line completely, so check by substituting $x = 0$.

Example: Find the slope and y -intercept of $y = 6x + 11$.

Step 1. Two different questions are being asked, so answer them with two different moves: one about a change, one about a single value.

Step 2. Change: going from x to $x + 1$ adds 6 to y . Value at zero: $6(0) + 11 = 11$.
Slope $m = 6$, y -intercept $b = 11$

Try it: Find the slope and y -intercept of $y = 4x + 9$.

check: $4 = m$ and $9 = b$

(!) **Watch out:** $y = 5x + 2$ and $y = 2x + 5$ use the same two numbers and are different lines. Never read a pair of numbers without asking which one is multiplying.

Skill 2 — Building the equation from two points

Rule: the slope is the change in y over the change in x . When one of the points sits on the y -axis, its y -value is b — it goes on its own, not in front of x .

Example: A line passes through $(0, 7)$ and $(4, 23)$. Find its slope.

Step 1. One point already has $x = 0$, so its height is the intercept; the slope still has to come from the change between the two points.

Step 2. $\frac{23 - 7}{4 - 0} = \frac{16}{4}$, so the equation is $y = 4x + 7$.

The 7 stays alone; the 4 multiplies. $y = 4x + 7$

Try it: A line passes through $(0, 5)$ and $(3, 20)$. Write its equation.

check: $5 = b$ and $20 = 5 + 3 \cdot 5$

Skill 3 — Building the equation from a table

Rule: the row where the input is 0 gives b ; the step from one row to the next, divided by the step in the input, gives m .

Example: Find m and b for this table.

x	0	1	2	3
y	8	13	18	23

Step 1. The table already contains the input value 0, so read b straight off and get m from the change across the whole table.

Step 2. At $x = 0$ the output is 8. From the first column to the last, $\frac{23 - 8}{3 - 0} = \frac{15}{3}$.

$$b = 8 \quad \text{and} \quad m = 5$$

Try it: A table gives $x = 0, 2, 4$ with $y = 12, 20, 28$. Find b and m .

$b = 12$ and $m = 4$:check